

• EXPLORE • DEFINE • EVALUATE • DEVELOP

CORPORATE PROFILE

- ✓ SAMREC-Compliant Reporting
- ✓ Advanced Technical Software-
Leapfrog Geo | Leapfrog Edge | Deswik | ArcGIS Pro
- ✓ ISO-Certified Management System
ISO 9001:2015 | ISO 14001:2015 | ISO 45001:2018



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COMPANY OVERVIEW

GEMAD (Geological Evaluation & Mineral Assets Development) is a South African Mineral Asset Development Entity focused on identifying, evaluating, developing and advancing high-potential mineral opportunities into technically sound and economically viable mining projects.

GEMAD integrates geology, exploration, mineral resource evaluation, mine design, mine planning, technical studies and project development into a single multidisciplinary solution.

Our approach enables clients, investors and project owners to move beyond the identification of mineral resources towards the development of investment-ready mineral assets.

From early-stage geological evaluation and exploration through resource definition, technical evaluation, mine planning and project development, GEMAD provides the technical framework required to make informed investment and development decisions.

GEMAD's mineral asset development philosophy is built around:

**Discovery → Geological Evaluation → Resource Definition
→ Technical Evaluation → Bankable Mineral Asset**

This integrated approach differentiates GEMAD from conventional service providers by connecting geological understanding with mining engineering, economic evaluation and practical project development.



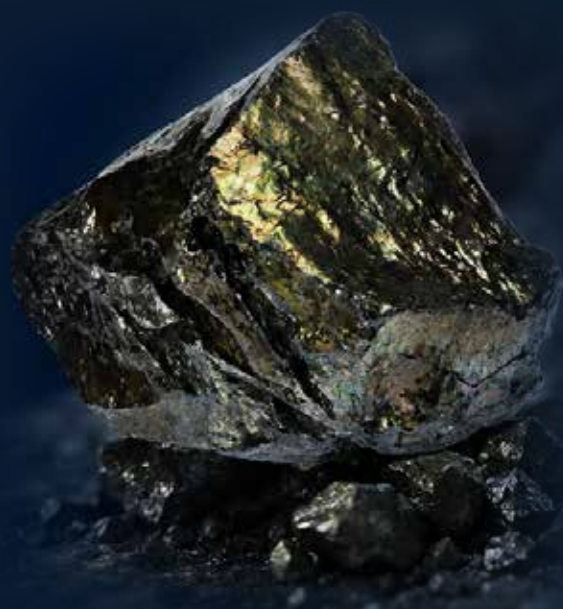
VISION

To become **Africa's leading** Mineral Asset Development Entity recognised for technical excellence, responsible resource development, and sustainable value creation.



MISSION

To identify and develop **high-potential mineral assets** through innovative geological evaluation, disciplined project development, and responsible mining practices that create long-term value for stakeholders, investors, and communities.



OUR VALUE PROPOSITION

*We don't just find resources
— we help develop assets.*

GEMAD combines scientific expertise, modern technology and mining industry experience to provide an integrated pathway from geological opportunity to investment-ready mineral project.

Our value proposition is built on five pillars:

GEOLOGICAL EXCELLENCE

Rigorous geological investigation, exploration, modelling and mineral resource evaluation

INTEGRATED MINING SOLUTIONS

Connecting geology with mine design, planning, scheduling and technical project development.

DATA-DRIVEN DECISION MAKING

Using industry-leading geological, GIS and mine planning technologies to improve the quality of technical decisions.

INVESTMENT READINESS

Developing technically robust studies and evaluations that support investment, funding and development decisions.

RESPONSIBLE MINERAL DEVELOPMENT

Integrating environmental, social, health, safety and community considerations into mineral asset development.





OUR CORE SERVICES

Geological Evaluation & Exploration

GEMAD provides geological services across the mineral asset development cycle, from early-stage desktop evaluation through exploration and resource definition.

Mineral Resource Modelling & Estimation

GEMAD converts geological information into reliable three-dimensional geological models and mineral resource estimates that support technical and investment decisions.

Services include:

- ▶ Geological desktop studies
- ▶ Geological mapping
- ▶ Target generation and delineation
- ▶ Exploration programme design
- ▶ Exploration supervision
- ▶ Exploration drilling supervision
- ▶ Core logging and sampling
- ▶ Geological data management
- ▶ QA/QC implementation
- ▶ Geological interpretation
- ▶ Exploration reporting
- ▶ Geological due diligence

Services include:

- ▶ Geological database evaluation
- ▶ Data validation and QA/QC
- ▶ 3D geological modelling
- ▶ Geological domain interpretation
- ▶ Grade estimation
- ▶ Block modelling
- ▶ Mineral Resource estimation
- ▶ Resource classification
- ▶ Geostatistical evaluation
- ▶ Resource reconciliation
- ▶ SAMREC technical reporting
- ▶ Competent Person technical support

Our modelling workflow utilises advanced geological and spatial technologies, including Leapfrog Geo, Leapfrog Edge and ArcGIS Pro.





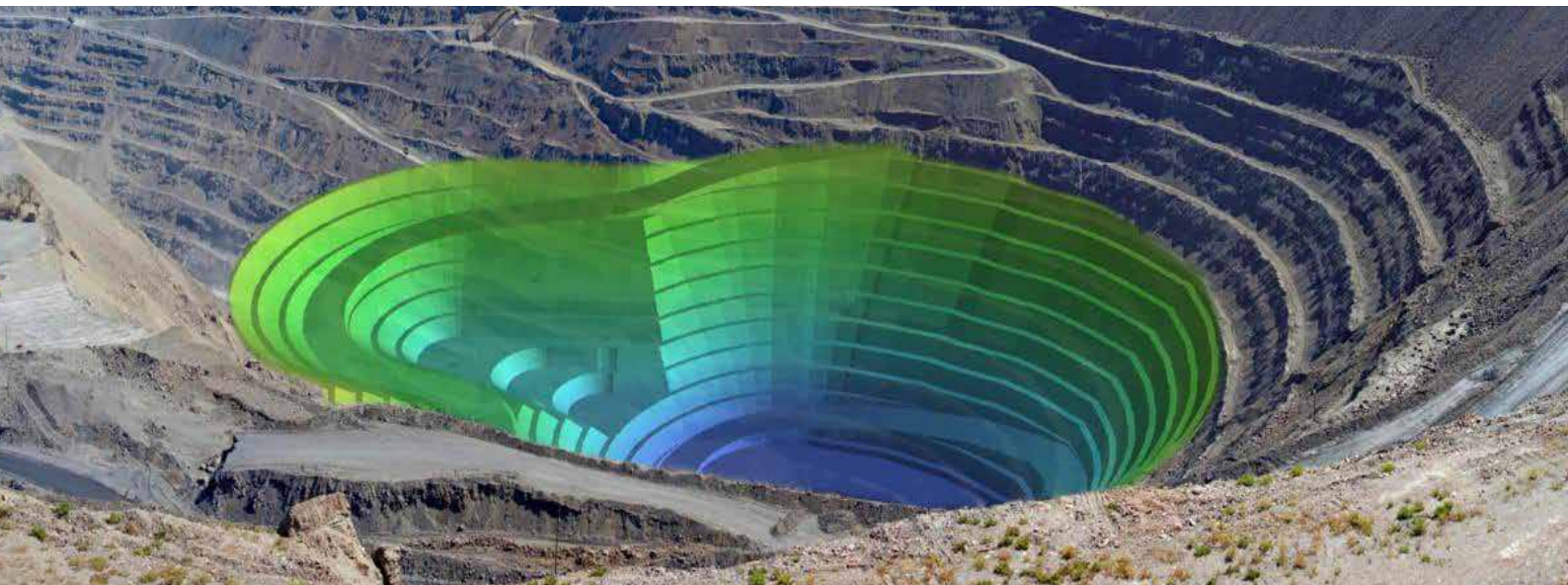
MINE DESIGN, PLANNING & SCHEDULING

GEMAD bridges the gap between a defined mineral resource and the practical development of a mine.

Our mining studies consider the geological characteristics of the deposit together with mining methods, production requirements, infrastructure and economic considerations.

Services include:

- ▶ Open-pit mine design
- ▶ Underground mine planning support
- ▶ Pit optimisation
- ▶ Pushback design
- ▶ Haul-road design
- ▶ Dump and stockpile design
- ▶ Mine infrastructure planning
- ▶ Production planning
- ▶ Life-of-mine planning
- ▶ Short-, medium- and long-term scheduling
- ▶ Equipment and fleet considerations
- ▶ Mining sequence optimisation
- ▶ Production scheduling
- ▶ Mine plan optimisation



GEMAD utilises **Deswik** mine design, planning and scheduling technology to develop practical and integrated mining solutions.

MINE PROJECT DEVELOPMENT

GEMAD supports mineral projects through the technical development stages required to determine their viability and investment potential.

Our studies include:

- | | |
|-------------------------------------|--|
| ● Conceptual studies >>> | ● Infrastructure assessment >>> |
| ● Scoping Studies >>> | ● Mining method evaluation >>> |
| ● Pre-Feasibility Studies >>> | ● Production strategy >>> |
| ● Bankable Feasibility Studies >>> | ● Project optimisation >>> |
| ● Technical due diligence >>> | ● Technical risk assessment >>> |
| ● Independent technical reviews >>> | ● Mineral asset valuation support >>> |
| ● Mine design and planning >>> | ● Investment-readiness assessments >>> |
| ● Mine scheduling >>> | ● Mine Financial Valuation >>> |

Our objective is to provide decision-makers with a clear understanding of the **technical, operational and economic potential of a mineral asset.**

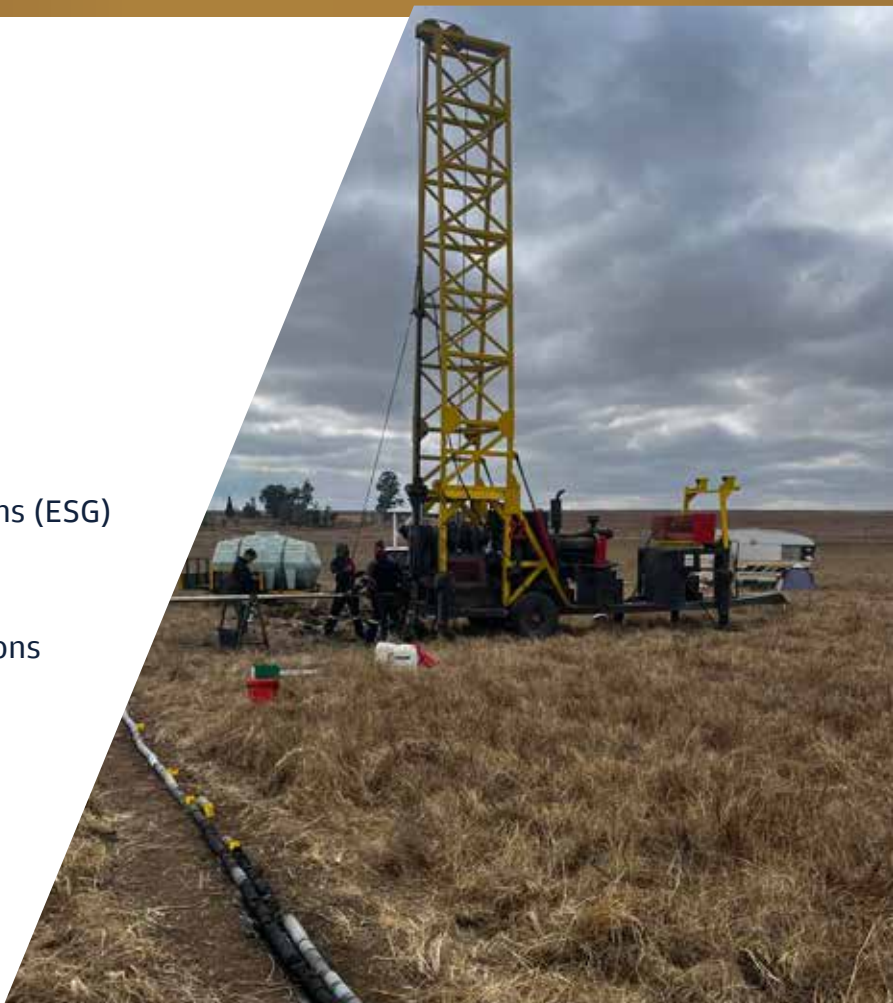
TECHNICAL DUE DILIGENCE & MINERAL ASSET EVALUATION

GEMAD assists investors, mining companies, financial institutions and project developers in evaluating mineral opportunities before significant capital is committed.

Our technical evaluation considers:

- Geological confidence
- Exploration data quality
- Mineral resource potential
- Resource classification
- Mining method
- Mineability
- Production potential
- Infrastructure
- Processing considerations
- Environmental and social considerations (ESG)
- Technical risks
- Development requirements
- Capital and operating cost considerations
- Project economics
- Development timelines

This enables stakeholders to distinguish between a **mineral occurrence**, a **mineral resource** and a **genuinely developable mineral asset.**



Modern mineral asset development depends on the quality of geological and operational data.

GEMAD uses advanced industry software to integrate geological, spatial and mining information into a coherent technical workflow.

TECHNOLOGY & DIGITAL MINING

**Our
technology
ecosystem
includes:**

ArcGIS Pro

GIS, spatial analysis, geological mapping, exploration targeting and geographic data management.

Leapfrog Geo

3D geological modelling, structural interpretation and geological visualisation.

Leapfrog Edge

Geostatistics, grade estimation and mineral resource evaluation.

Deswik

Mine design, planning, scheduling and integrated mining studies.

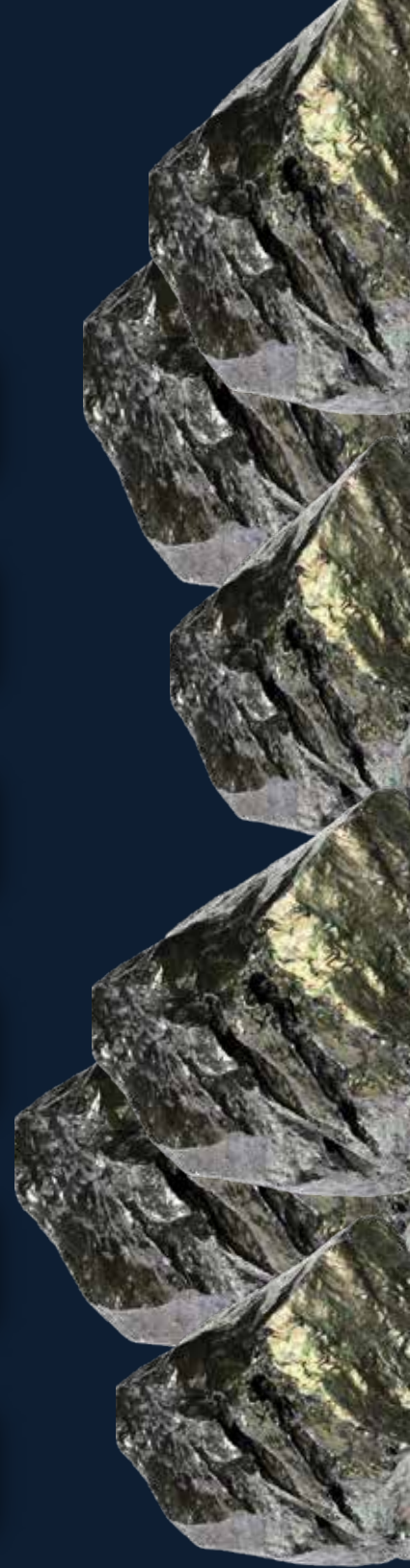
The integration of these platforms allows GEMAD to establish a workflow from:

SPATIAL DATA → GEOLOGICAL INTERPRETATION → 3D MODEL → RESOURCE MODEL → MINE DESIGN → MINE SCHEDULE → PROJECT EVALUATION

This integrated digital workflow improves technical consistency and supports data-driven decision-making.



OUR MINERAL ASSET DEVELOPMENT PROCESS



COMMODITIES



GEMAD brings together professionals with multidisciplinary experience across geology, mining, exploration, engineering, environmental management and mineral project development.

OUR
TECHNICAL TEAM

OUR TECHNICAL CAPABILITY INCLUDES:

- Professional Geologists
- Exploration Geologists
- Resource Geologists
- Mining Engineers
- Mine Planning Specialists
- Geological Modellers
- Environmental Specialists
- Engineering Professionals
- Project Development Specialists
- GIS specialists

The team combines academic knowledge, professional registration, practical mining experience and modern technical software capability.



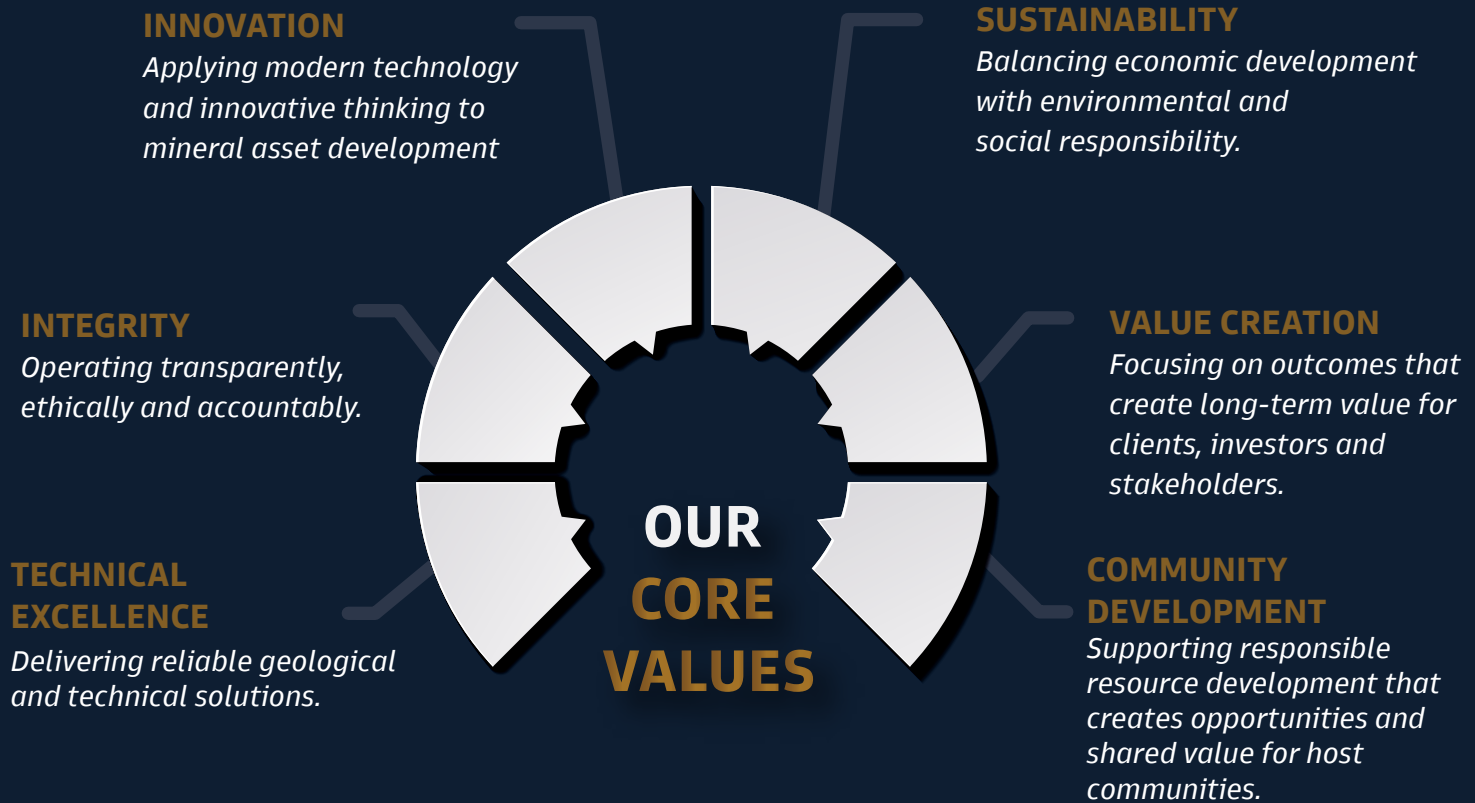
— QUALITY, HEALTH, SAFETY & ENVIRONMENT —

GEMAD is committed to delivering services through a structured Integrated Management System.

Our management system incorporates:

ISO 14001:2015	ISO 9001:2015	ISO 45001:2018
Environmental Management	Quality Management	Occupational Health & Safety
Supporting responsible environmental management throughout project activities.	Ensuring consistent processes, quality control and continuous improvement.	Promoting safe working environments and effective management of occupational health and safety risks.
Our commitment to quality extends beyond compliance. We seek to establish technical processes that are repeatable, auditable and capable of producing consistent results.		





WHO WE SERVE

GEMAD provides technical support to a diverse range of stakeholders within the mineral resources sector.

Mining Companies

Technical support for exploration, resource development, mine planning and project optimisation.

Junior Exploration Companies

Technical capability to advance exploration projects and strengthen mineral asset value.

Investors & Private Equity

Independent technical evaluation and due diligence of mineral opportunities.

Funding Institutions

Technical assessment and support in evaluating the viability and development potential of mining projects.

Mineral Rights Holders

Technical evaluation and development strategies for mineral assets.

Project Developers

Integrated geological and mining solutions from exploration through feasibility.

Government & Development Institutions

Technical support for mineral development programmes, project evaluation and responsible resource development.

GEMAD AS A MINERAL ASSET DEVELOPMENT PARTNER

GEMAD's role extends beyond providing individual technical services.

We aim to become a **long-term technical and strategic partner** to mineral asset owners and developers.

Our integrated model allows a client to access multiple technical capabilities through one coordinated platform.

Instead of treating geology, resource modelling, mine planning and project evaluation as isolated activities, GEMAD connects them into a single development pathway.



Our
integrated
model:

↓
GEOLOGICAL EXPLORATION

↓
RESOURCE MODELLING
AND ESTIMATION

↓
MINE DESIGN

↓
MINE PLAN

↓
SCHEDULE

↓
ECONOMIC EVALUATION

↓
PROJECT DEVELOPMENT

↓
BANKABLE MINERAL ASSET

SOUTH AFRICA & AFRICA

GEMAD is headquartered in South Africa and is positioned to serve mineral projects across the African continent.

Our strategy is to develop a regional footprint within the **SADC mining market**, with particular emphasis on countries possessing significant geological potential and opportunities for responsible mineral asset development.

GEMAD has expanded its presence into Botswana through **GEMAD Botswana**, creating a platform from which to engage with Botswana's mining and exploration sector.



Our broader African strategy is centred on building partnerships with:

- Mining companies
- Junior explorers
- Mineral-rights holders
- Investors
- Development finance institutions
- Government institutions
- Local communities
- Technical partners

INTEGRATED GEOLOGY-MINING SOLUTIONS

We connect geological evaluation with mining engineering and project development.

INDUSTRY-LEADING SOFTWARE

Our technical workflow incorporates ArcGIS Pro, Leapfrog Geo, Leapfrog Edge and Deswik.

SAMREC-ALIGNED TECHNICAL CAPABILITY

Our resource evaluation and reporting approach is structured around recognised mineral reporting standards.

MULTIDISCIPLINARY TEAM

Our professionals combine geology, mining, engineering, environmental and project development capabilities.

ISO-CERTIFIED MANAGEMENT SYSTEM

Our quality, environmental and occupational health and safety systems are aligned with ISO 9001, ISO 14001 and ISO 45001.

INVESTMENT-ORIENTED APPROACH

We focus on progressing mineral opportunities towards technically robust and economically viable projects.

AFRICAN MARKET KNOWLEDGE

Our South African base and expanding SADC presence position GEMAD to support mineral development across Africa.

WHY
GEMAD
?

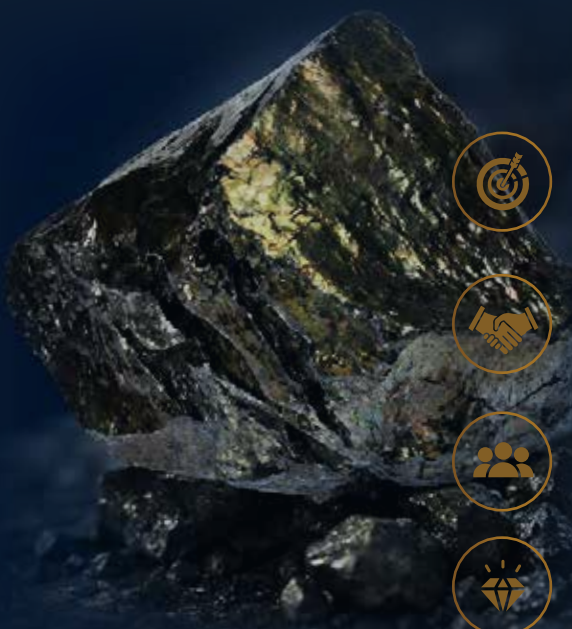
OUR COMMITMENT

GEMAD believes that Africa's mineral wealth represents an opportunity not only for resource extraction, but **for long-term economic development, industrialisation, investment and community value creation.**

Our commitment is to apply sound geological science, modern technology, engineering expertise and responsible development principles to every mineral opportunity we evaluate.

We aim to help our clients answer the most important question:
Can this mineral opportunity become a viable, investable and developable mining project?

GEMAD exists to help provide that answer.



PREVIOUS EXPERIENCE AND REFERENCES

Project Name	Company / Client	Commodity / Sector	Status	Period
TNW Core Drilling – Exploration, Manungu Colliery	Mbuyelo Coal	Coal	Completed	March 2019 – May 2019
Detailed Exploration Project & Technical Report	Benk Holdings	Coal	Completed	March 2021 – May 2021
TNW Core Drilling – Exploration, Ntshovelo Colliery	Mbuyelo Coal	Coal	Completed	March 2022 – October 2023
Water Exploration Project, Khashani Colliery	Mbuyelo Coal	Water / Mining Support	Completed	January 2023 – February 2023
TNW Core Drilling – Exploration, Khashani Colliery	Mbuyelo Coal	Coal	Completed	May 2023 – June 2023
Percussion Drilling, Khashani Colliery	Mbuyelo Coal	Coal Exploration	Completed	August 2023 – September 2023
Geological Desktop Study – QAMU 17486 GT & Uithoek 1078 GT	Confidential / Client Reference Available	Geological Evaluation	Completed	—
Geotechnical Investigation – Opencast Mine Haul Road	Arnot Opco	Coal / Geotechnical	Completed	—
Geological Services & Mine Planning	Benk Colliery	Coal	Ongoing	—
Lettieskeus Exploration	Eloh Energy	Coal	Ongoing	



GEMAD Project Experience

GEMAD's project portfolio demonstrates practical experience across:

- Coal exploration and geological investigation
- TNW core drilling
- Percussion drilling
- Detailed exploration programmes
- Geological desktop studies
- Technical report compilation
- Water exploration
- Geotechnical investigations
- Mine planning and geological mining support
- Mining infrastructure support

This project experience demonstrates GEMAD's ability to provide integrated geological, exploration, mining and operational support services across different stages of the mineral asset development cycle.

MANAGEMENT TEAM



MANAGING DIRECTOR

Gijimani Motau

BSc (Hons) Geology | Pr.Sci.Nat

Gijimani Motau is an experienced geologist and aspiring mining entrepreneur with extensive expertise in mineral exploration, mining operations, and resource development. He specialises in exploration, geological modelling, and resource estimation, with strong experience in managing exploration and drilling programmes, as well as project development. His business interests focus on identifying, evaluating, and advancing high-potential mineral projects into economically viable mining operations. He has a keen interest in coal, Platinum Group Metals (PGMs), and base metals.



EXPLORATION GEOLOGIST

Isaac Mabonda

BSc (Hons) Geology |
Pr.Sci.Nat

Over 14 years of experience in open-cast mining, production geology, and greenfield exploration projects. Skilled in geological control, grade management, drill supervision, geological modelling, and exploration programme management across commodities including chromite, PGE, coal, iron ore, gold, nickel, and manganese. Experienced in both South African and international mining environments.

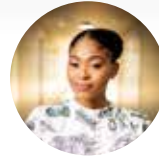


ENVIRONMENTAL CONSULTANT

Njabulo Mxusile

BSc (Hons) Environmental
Management | Pr.Sci.Nat

More than 5 years of experience in environmental management within the diamond and coal mining industries. Environmental Consultant and Professional Natural Scientist specializing in environmental compliance, regulatory management, and sustainable mining practices. Skilled in environmental auditing, environmental authorisations, water management, waste management, rehabilitation planning, and ISO 14001:2015 compliance across underground and surface mining operations.



RESOURCE GEOLOGIST

Nangamso Magobiyane

BSc (Hons) Geology |
Pr.Sci.Nat | MBA

Over 15 years of experience in mining and mineral exploration. Professional Natural Scientist and Senior Geologist specializing in geological modelling, mine planning, operational geology, and mineral resource estimation across underground and surface mining operations. Experienced in managing drilling programmes, multidisciplinary technical teams, and integrating geological, geochemical, and geophysical data to support strategic project decisions from exploration through to production. Affiliated with SACNASP, SAIMM, CGSSA, and WIMSA.



JUNIOR MINE PLANNER

Tlotlo Kearata Poti

BEng Mining Engineering

Tlotlo is a Mining Engineer with over three years of experience in the mining and construction sectors. His expertise spans mine planning and design, production scheduling, drill and blast engineering, costing, explosive handling, geotechnical analysis, soil profiling, and mining design simulations. Based in Botswana, he supports the delivery of technically sound and efficient mining solutions across the project development lifecycle.

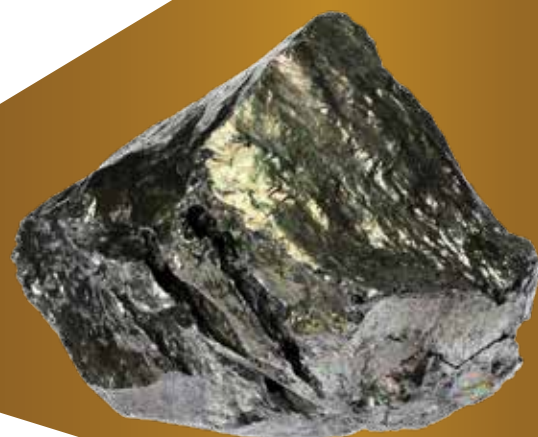


CORPORATE POSITIONING



Geological Evaluation & Mineral Assets Development
"From discovery to bankable mineral assets"

Integrated Geology – Mining – Project Development



Contact GEMAD

Ready to Advance Your Mineral Project?

Whether you are an **investor, mining company, mineral-rights holder, junior explorer or project developer**, GEMAD provides the technical expertise required to unlock the value of mineral assets.

GEMAD

Turning Mineral Resources into Bankable Mineral Assets

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